

RippleLogic Cascade Standard v1.0

Source-boundary rule: If this compact standard conflicts with the RippleLogic Canon, the Canon controls.

Release: MathGov Core Release 2026.09 v11.4.6 Final GitHub Readiness Patch

This standard is the compact public reference for the current five-level RippleLogic cascade inside the MathGov framework. It prevents companion documents from repeating long synchronization notes.

TRC-CSV feedback rule. If CSV discovers a catastrophic, irreversible, or ruin-path scenario not represented in TRC, the run MUST reopen TRC before RLS. CSV does not absorb TRC.

Gate-critical confidence guard. Low confidence in adverse rights-covered, catastrophe-covered, or material CSV impacts cannot by itself make an option pass RF/NCRC, TRC, or CSV. If a gate outcome could change, the run must use a governed conservative bound, collect evidence and rerun, narrow the claim, downgrade, escalate, or refuse the stronger claim.

Baseline pointer. Gate-admissibility cells and residual welfare cells may use distinct baselines where the Canon requires it; see Canon §5.1A for the floor-reference versus status-quo dual-baseline rule.

Public cascade

RG -> RF -> TRC -> CSV -> RLS

Plain language: Ground reality -> protect rights -> bound ruin -> preserve the structure -> score the ripples.

Formal shorthand

RG/RSG -> RF/NCRC -> TRC -> CSV -> RLS

The formal components are:

- RG/RSG: Reality Grounding / Reality-Surface Grounding. Claim-authority precondition, not an ordinary option-rejecting gate.

RG status rule: Reality Grounding is Level 1 of the public method and can force narrowing, escalation, exploratory marking, or refusal. RF/NCRC, TRC, and CSV are the option-filtering gates; RLS ranks only what survives them.

RF/NCRC: Rights Floor, formally implemented by the Non-Compensatory Rights Constraint. Non-compensatory gate.

TRC: Tail-Risk Constraint. Non-compensatory catastrophic-risk gate.

CSV: Containment and Structural Viability. Selectability gate for containing-system integrity and execution viability.

RLS: RippleLogic Score. Residual welfare-ranking layer applied only to options that remain selectable.

Computability vs realizability rule

A computed, simulated, generated, or model-fluent option is not automatically grounded, selectable, executable, or ethical. RG supplies claim authority; RF/NCRC supplies rights admissibility; TRC supplies ruin bounding; CSV supplies structural and execution viability; RLS ranks only the surviving selectable set.

Source-coupling rule

A downstream output, interface result, benchmark success, compliance status, institutional permission, inherited procedure, or model-fluent answer is not proof of grounded capability. When material, Reality Grounding must record the claimed capability, enabling conditions, boundary conditions, source evidence, inherited assumptions, downstream compensations, and required claim action. CSV consumes this evidence when weak source coupling creates dependency, containment, or structural-viability risk.

Source-Coupling Integrity is not a sixth gate. It is a Reality Grounding subdiscipline and a CSV diagnostic where material.

Non-overlap rule

RG determines what claim may be made. RF/NCRC, TRC, and CSV determine whether an option remains selectable. RLS ranks only selectable options. RLS cannot rescue rights-floor failure, tail-risk failure, or CSV failure.

CSV non-purity rule

CSV does not reject every negative ripple. Negative ripples must be made visible and routed. Bounded residual harms may enter RLS. Uncontained, structurally degrading, unjustly externalized, non-viable, hidden, lock-in-producing, unmonitored, or beyond-tolerance harms require controls, redesign, escalation, emergency-provisional handling, or failure.

All-Encompassing Infinite Union (AIU) and SGP boundary

All-Encompassing Infinite Union (AIU) is a horizon/meta-union orientation, not a Tier 1-3 scoring object or override. SGP is a moral-status and protection evidence interface; it informs protected-stakeholder modeling where permitted but does not replace RG, RF/NCRC, TRC, CSV, RLS, lawful authority, or governance-role requirements.